

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	HDI Prediction Engine	Accepts Life Expectancy, Mean Years of Schooling, GNI per capita, and Internet Users as input and predicts the HDI score using a trained Linear Regression model.	ML Model / Flask Backend	Done	1
2	HDI Category Classification	Classifies the predicted HDI score into Low, Medium, High, or Very High HDI category and displays it to the user.	Flask Backend (app.py)	Done	0.5
3	Data Preprocessing & Cleaning	Handles missing values, removes non-numeric columns, and prepares the dataset for model training using Pandas.	Training / Jupyter Notebook	Done	0.5
4	Model Persistence (Pickle)	Saves the trained Linear Regression model using Pickle so it can be reused for predictions without retraining.	Training Module	Done	0.5
5	Web-Based User Interface	Provides a simple, responsive home page and input form for users to enter indicators and view results.	Flask Templates (HTML/CSS)	Done	0.5
6	HDI Score Comparison Dashboard	Displays Top 20 and Bottom 10 countries by HDI score using interactive bar charts for quick comparison.	Dashboard (Chart.js)	Done	0.75
7	World HDI Map Dashboard	Shows country-wise HDI data as filterable cards by category, with Pie and Doughnut chart visualizations.	Dashboard (Chart.js)	Done	0.75
8	HDI Trend Over Years Dashboard	Visualizes HDI trends from 1990–2015 by region using a line chart, with a radar chart for multi-indicator comparison.	Dashboard (Chart.js)	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Prediction Engine, Classification, Preprocessing, Model Persistence, Web UI)
Additional / Nice-to-Have Features	3 (HDI Comparison Dashboard, World HDI Map Dashboard, HDI Trend Dashboard)
Features Tested & Verified	8 / 8 — All features tested and working as expected

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page, Input form, Result page
2	Backend / Logic	2	HDI prediction logic, Category classification
3	Database / Storage	1	Model persistence using Pickle (HDI.pkl)
4	API / Integration	1	Kaggle dataset integration for training data
5	Security / Authentication	0	No authentication required (open access local app)